

LOGAN LAPIERRE

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OBJECTIVE / SUMMARY

UW–Milwaukee CS senior building agentic workflows, real-time voice agents, and local-first AI applications. Combining AI software engineering with a strong Linux, Docker, C/Python backend, and cloud pipeline foundation. Seeking AI Engineering, Applied AI, or Agentic AI internships.

EDUCATION & CERTIFICATIONS

University of Wisconsin–Milwaukee | B.S. in Computer Science (Expected Dec 2026 • Full-time Jan 2027)

- **Relevant Coursework:** Systems Programming, Operating Systems, Server-Side Internet Programming, Computer Architecture, Data Structures & Algorithms, Software Engineering.
- **Certifications & Training:** AWS Cloud Practitioner Essentials • Google IT Automation with Python • LFCS (Linux Foundation Cert System Admin — In Progress)

TECHNICAL SKILLS

- **AI & Agentic Systems:** LiveKit Agents, Deepgram STT, ElevenLabs TTS, Silero VAD, WebRTC, Agentic Scrapers (Apify API), Playwright Automation, LLM Reranking, Prompt Engineering, Guardrails.
- **Local & On-Device AI:** Hugging Face Transformers (transformers, torch), Ollama API, openWakeWord, pytsx3, Local Model Memory Caching (FLAN-T5, TinyLlama, Phi-2).
- **Software Engineering & Infrastructure:** Python, C, C++, JavaScript, SQL, Bash, Next.js, React 19, Flask, FastAPI, Django, PostgreSQL, Docker, AWS (ECS, EC2), Terraform, GitHub Actions CI/CD, Git.

AI ENGINEERING PROJECTS

LiveKit AI Mock Interview Agent | *Python, LiveKit, Groq API, Deepgram, ElevenLabs, Next.js, Neon Postgres*

- Built a real-time voice AI interviewer using LiveKit Agents, Deepgram STT, ElevenLabs TTS, and Groq (Llama-3.3-70b) via OpenAI SDKs to manage turn handling and low-latency WebRTC streaming.
- Architected a 2-stage conversational state machine (SelfIntroductionAgent -> PastExperienceAgent) with function tools and fallback timers (`_timeout_watcher`). Calibrated endpointing (2–6s delay) to accommodate candidate pauses.
- Engineered an offline Python unittest suite (`tests/test_agent.py`) using in-process LiveKit fakes to validate state handoffs without consuming API credits.

Autonomous AI Job Source Agent | *Python, Apify API, Playwright, Gemini 1.5 Flash, Pytest*

- Architected an agentic web crawler extracting corporate domains from LinkedIn URLs via Apify API and pinpointing active career portals using Gemini 1.5 Flash link reranking.
- Implemented Playwright Chromium headless scripts (`crawler.py`) to navigate dynamic JavaScript single-page apps and harvest direct job posting URLs.
- Engineered a resilient fallback workflow with domain validation (`validator.py`), retries, and rule-based keyword link scoring when LLM keys or quotas are absent.
- Authored a Pytest suite featuring mock execution flags (`--mock`), enabling deterministic offline test verification without hitting external APIs.

Local AI Chatbot | *Python, Flask, Hugging Face, Ollama, openWakeWord, Electron, React*

- Developed a local-first, offline-capable desktop chat app combining a Python Flask backend (`ai_service.py`), Hugging Face Transformers, and Ollama to run open-weights models on-device.
- Designed a modular service layer supporting dynamic model loading and memory caching for local models (FLAN-T5, TinyLlama, Phi-2) and Ollama API endpoints (127.0.0.1).
- Integrated an on-device hands-free voice engine (`wake.py`) combining openWakeWord detection, PyAudio streaming, and pytsx3 offline TTS, packaged in Electron process wrappers.

OPEN SOURCE CONTRIBUTION

LMMS Digital Audio Workstation | *C++, Qt Framework, Git/GitHub ([PR #8074](#))*

- Authored a submitted open-source pull request proposing an experimental AudioRecorder plugin in C++ for the LMMS Digital Audio Workstation; engaged in community code review and technical feedback iteration.

EXPERIENCE

Server / Barista | New Perspective Senior Living — Waukesha, WI (*June 2026 – Present*)

- Fulfill daily dining room service, meal delivery, sanitation, and barista responsibilities for senior residents, coordinating service flow with culinary staff while maintaining operational reliability, clear communication, and adaptability alongside computer science studies.